

Oten Soft: Building a Secure Digital Future

KEY MANAGEMENT SYSTEM

Lucy, Zane



Agenda

- What is Key?
- The “Digital Key” - The Language of Trust in Cyberspace
- About KMS & The Fortress TEE
- The first KMS story - Google Workspace Data
- The practical challenges (Secret Management, Data Security ..)
- Into the Oten Ecosystem



What is an Encryption Key?

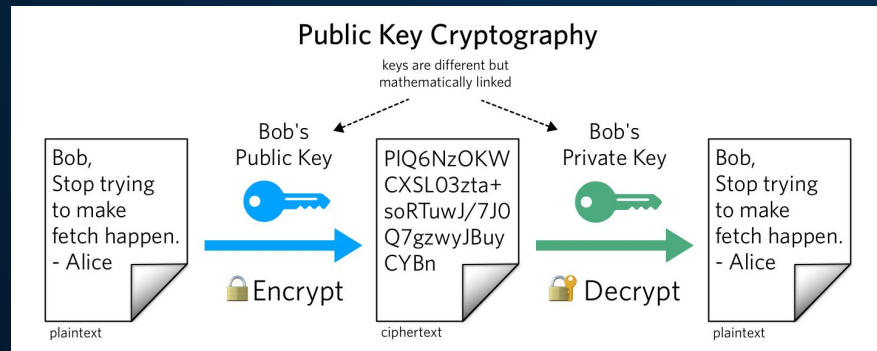
Image: A simple illustration of encryption and decryption.

[Plain Data] --(Encrypt with Key)--> [Scrambled Data]
[Scrambled Data] --(Decrypt with Key)--> [Plain Data]

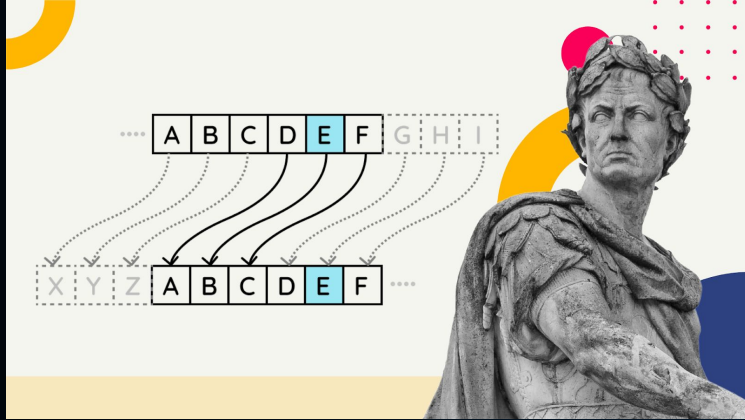
It is a special string of digital characters (bits), like a digital key.

What is it used for?

- The KEY is used to SCRAMBLE (encrypt) data into an unreadable format.
- The same KEY (or a key pair) is used to RESTORE (decrypt) the data to its original state.



Key history and development



100 BC

The Ceasar code



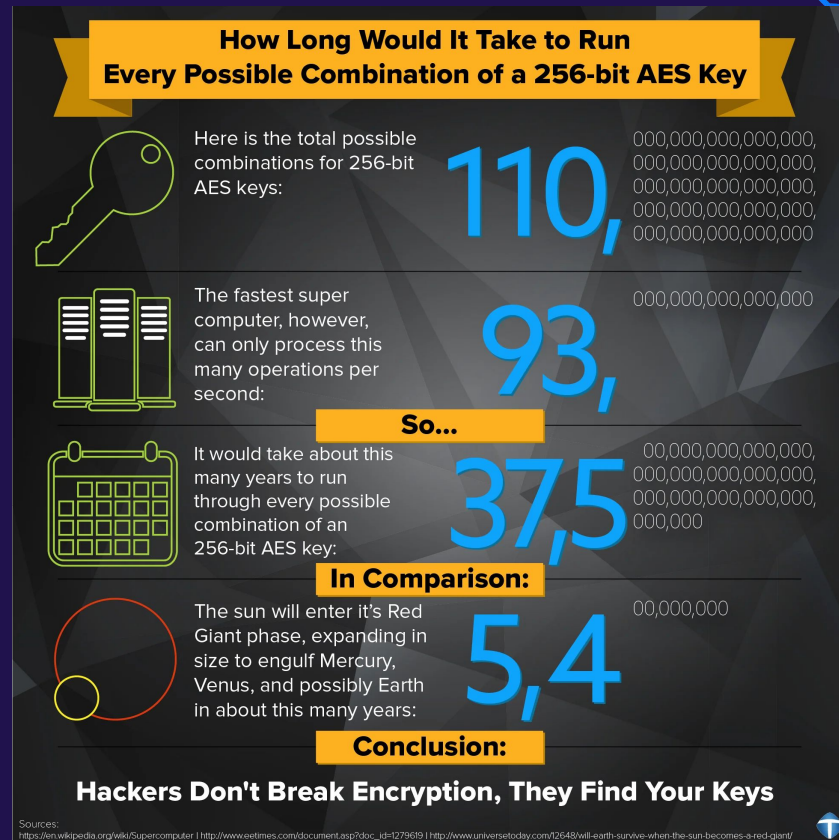
World War II
Enigma

1970s: The Data Encryption Standard (DES). This was one of the first encryption standards with a "key" to be approved by the US government.

By the late 1990s, advances in computing power had rendered the 56-bit key insecure, **proving that key length is a critical factor.**

- Key Length: 256 Bit - An AES-256 key is a string of 256 bits in a row.**
 - Example: 11001010... (a sequence of 256 zeros and ones).
 - **The total number of possibilities is: $2 \times 2 \times 2 \dots$ (256 times) = 2^{256}**

- Key Length: 256 Bit - An AES-256 key is a string of 256 bits in a row.**
 - Example: 11001010... (a sequence of 256 zeros and ones).
 - **The total number of possibilities is: $2 \times 2 \times 2 \dots$ (256 times) = 2^{256}**



Algorithms are used to generate keys?

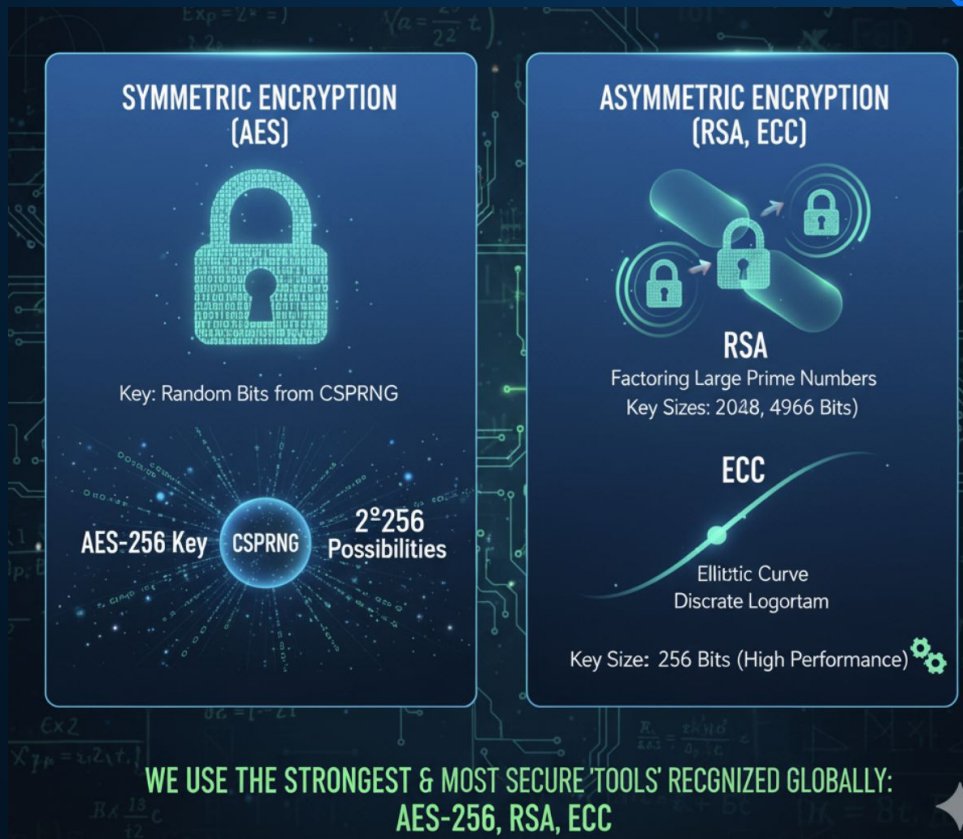


For Symmetric Encryption (AES): The key is a string of random bits. The core "algorithm" here is a Cryptographically Secure Pseudorandom Number Generator (CSPRNG). An AES-256 key is 256 bits long

For Asymmetric Encryption (RSA, ECC):

- RSA: Relies on the difficulty of factoring large numbers into their prime factors. Common key sizes are 2048 or 4096 bits.
- ECC (Elliptic-Curve Cryptography): ECC provides security equivalent to RSA but with much shorter key lengths (e.g., 256 bits), resulting in higher performance.

We don't invent our own algorithms. We use the strongest and most secure 'tools' recognized such as AES-256 and ECC.*



About KMS

The Digital Dream... But What About the Nightmares?

Hidden Dangers in Your Data

In digital world, your data is precious. KMS create value, but how do we truly keep everything safe from hidden dangers?

→ Key Chaos

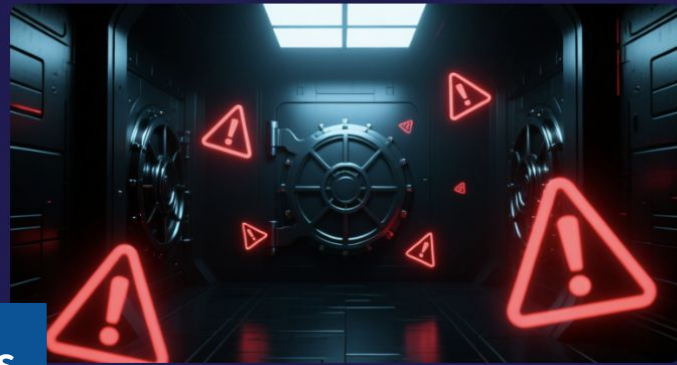
*Managing
hundreds of
encryption keys*

→ Lost key Risks

*If keys fall into wrong
hands*

→ Access Controls

*Who has the right to use
which key, and when?*



TEE - Trusted Execution Environment



A fully **isolated execution environment** inside a server, where sensitive data and critical code are processed with security guarantees far beyond a normal application.

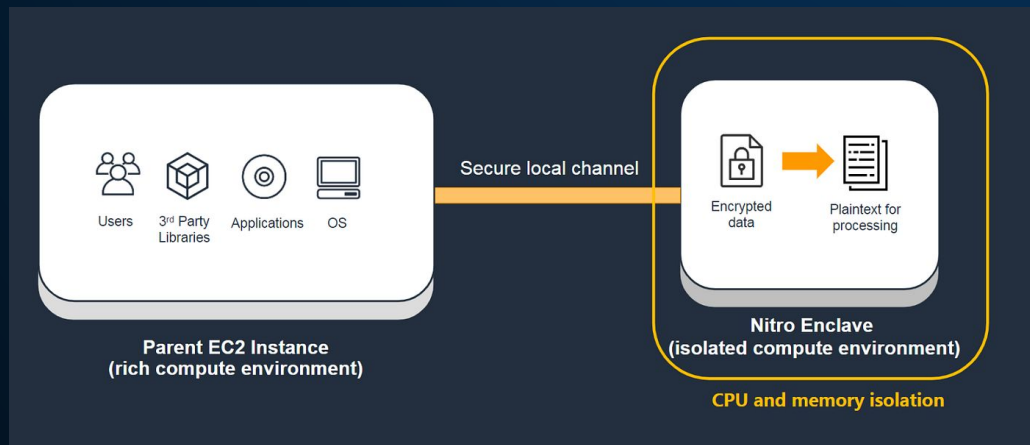
Hardware isolation: Dedicated CPU and RAM, OS, hypervisor, and parent application

No network: No IP address, no SSH, no inbound/outbound connectivity

No shell or filesystem: Runs only a single packaged application; no login, no file access

VSOCK-only communication: secure local channel

Attestation: Verifies the code running inside the TEE using cryptographic hashes and certificates, ensuring authenticity and preventing tampering



TEE - Trusted Execution Environment



1. Resistant even if OS, root, or hypervisor is compromised

Built on Nitro Hardware Isolation. OS cannot read enclave memory. OS cannot attach debuggers. OS cannot inject code.

2. Zero attack surface

NO Network. NO Shell. NO Filesystem

3. Code inside the TEE cannot be modified

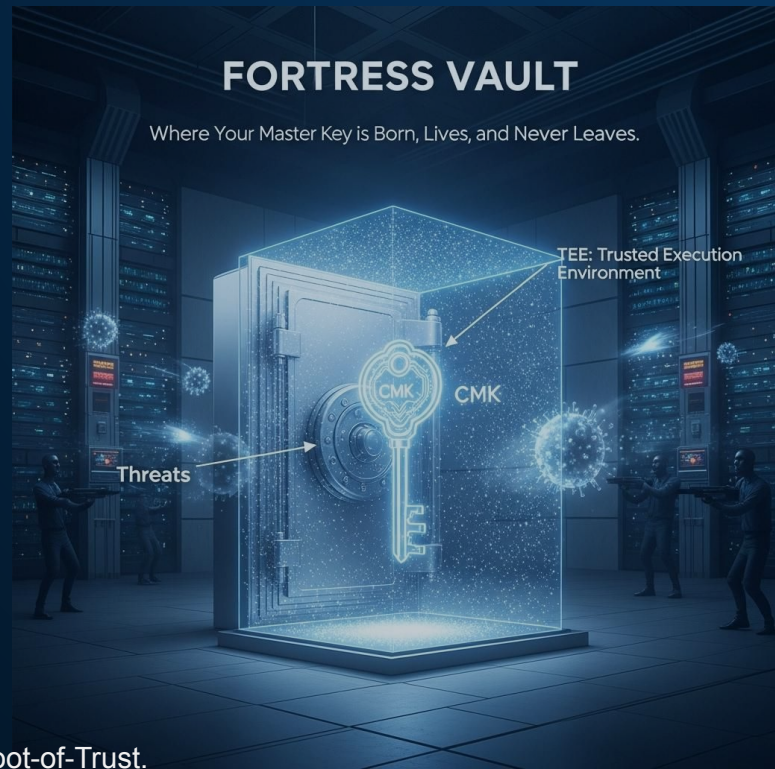
Attestation verifies: PCR0/1/2, PCR3/4, PCR8.

4. Sensitive data never leaves the enclave

Data exits ONLY when enclave explicitly sends output.

5. TRUE Zero-Trust Execution

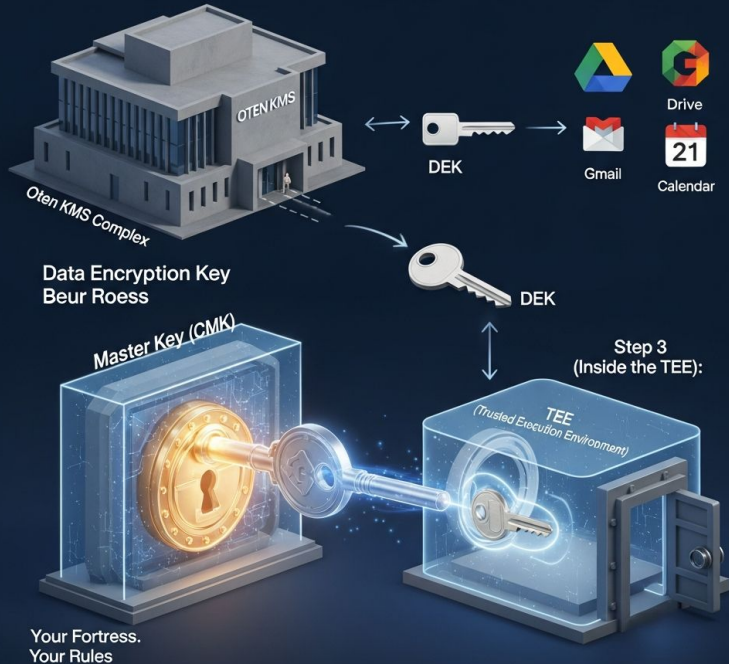
TEE trusts NO OS, NO Parent, NO Network — only Attested Code and AWS Root-of-Trust.



1st KMS Story - Google CSE



The Secure Bridge to Google CSE



Full Key Ownership: Your organization keeps complete control of all CMK/DEK keys. Google never has access to them.

Zero Data Exposure: Google stores the files but cannot decrypt them.

No Internal Leakage Risk: Developers and DevOps teams cannot view real keys.

Strong Compliance Alignment: Easily meets strict security standards like ISO 27001, SOC 2, GDPR, and PCI-DSS through secure key governance and auditable operations.

Case Study - Secret Management

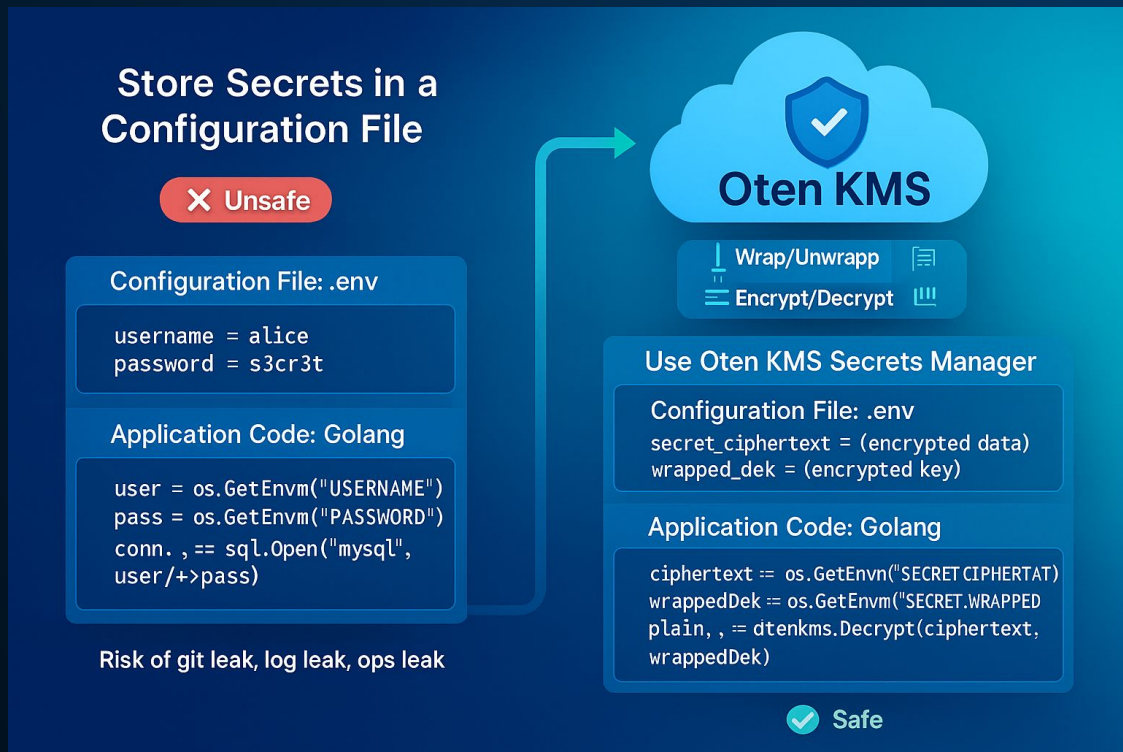


Secrets **never** appear in files, Git, logs, or configs

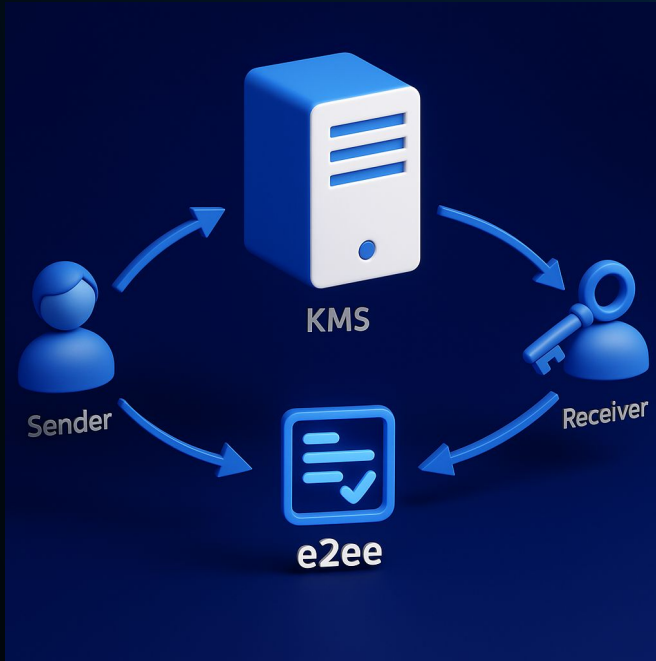
No engineer or operator can view real secrets

Full audit visibility: who accessed what and when

All secrets protected via DEK + CMK inside Oten KMS



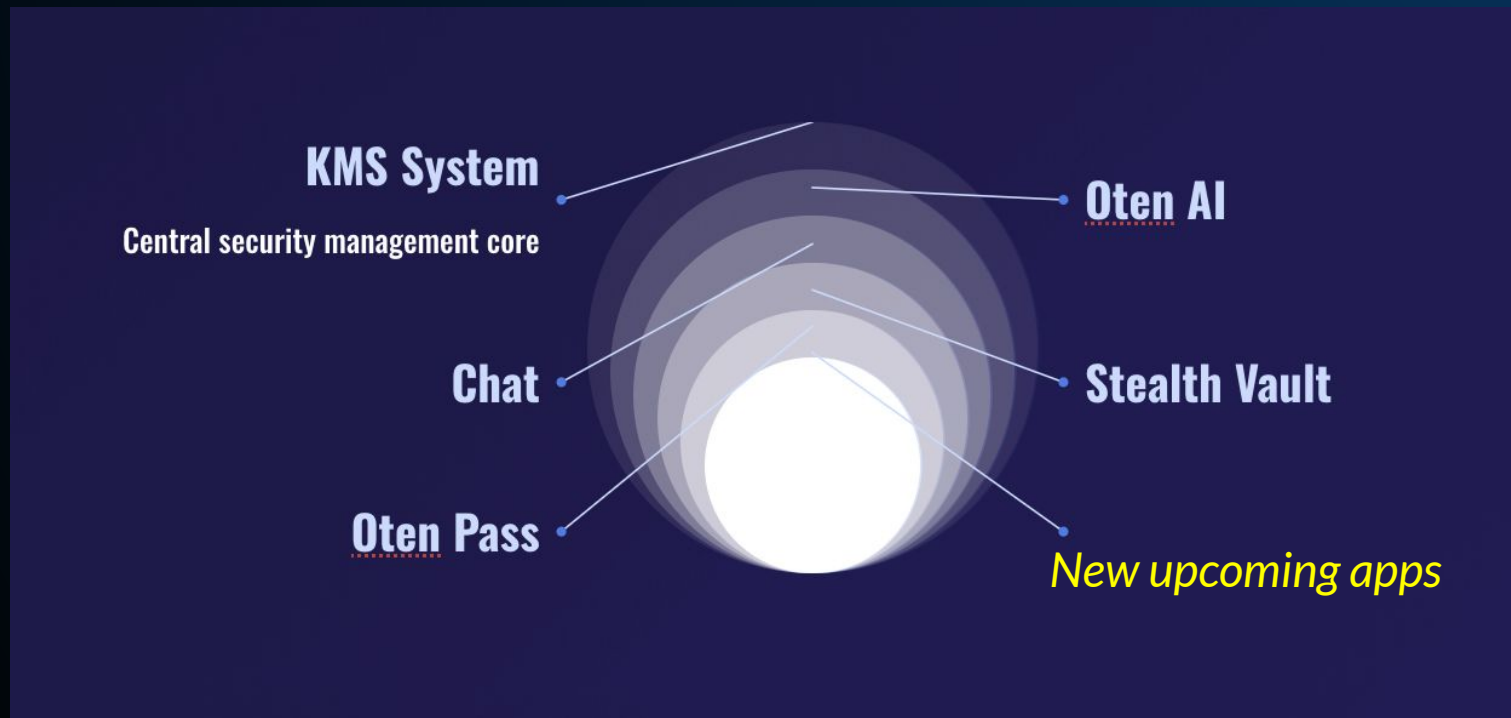
Case Study - Others



Column-level encryption
with Oten KMS



Oten Soft: Building a Secure Digital Future





THANKYOU

